

Yan Zeng

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EDUCATION

Technical University of Denmark (DTU)

MSc in Computer Science and Engineering GPA: 10.39/12

Lyngby, Denmark

Sept. 2024 – Jun. 2026 (expected)

Chalmers University of Technology (CTH)

Summer School

Gothenburg, Sweden

Jun. 2025 – Aug. 2025

Chongqing University of Posts and Telecommunications

B.Eng. in Software Engineering GPA: 3.40/4.0, Rank: 3/32

Chongqing, China

Sept. 2020 – Jun. 2024

RESEARCH EXPERIENCE

One-Step Visual Counterfactual Generation with Pixel Mean Flow

Supervisors: Prof. Anders Bjorholm Dahl, PhD Student Oskar Kristoffersen

Lyngby, Denmark

Feb. 2026 – Present

- Proposed a novel counterfactual generation framework based on Transformer architectures and Pixel Mean Flow, enabling one-step counterfactual generation through classifier-guided editing.
- Developed adaptive guidance and dynamic masking strategies to improve semantic consistency and editing precision while preserving high counterfactual validity.
- Demonstrated strong performance on two natural image datasets and one medical imaging dataset, achieving comparable or superior generation quality to state-of-the-art methods with up to $4\times$ faster inference. (Manuscript in preparation, targeting BMVC 2026)

3D Cardiac Image Synthesis with Flow Matching

Supervisor: Asst. Prof. Jiancheng Yang (ELLIS Institute Finland and Aalto University)

Helsinki, Finland(Remote)

Jan. 2026 – Present

- Designed a 3D cardiac image synthesis framework based on flow matching to generate realistic volumetric heart images with improved anatomical fidelity.
- Focused on enhancing anatomical structure preservation and generation quality to better model fine-grained cardiac morphology.
- Validated the generated data through downstream segmentation experiments, measuring the impact of synthetic samples on segmentation accuracy and robustness.

Counterfactual Medical Image Generation Using Latent Representations

Supervisors: Assoc. Prof. Anders Nymark Christensen, PhD Student Changlu Guo

Lyngby, Denmark

Sept. 2025 – Jan. 2026

- First extended classifier-guided counterfactual generation from pixel space to latent space, enabling a more efficient and scalable counterfactual generation framework.
- Redesigned the masking mechanism for latent-space counterfactual generation by replacing pixel-difference masks with gradient-based attribution masks, yielding more localized and semantically meaningful edits.
- Benchmarked the method on three medical imaging tasks, achieving high-quality counterfactual generation with substantial efficiency gains over prior classifier-guided diffusion baselines, including up to $40\times$ faster inference and $13\times$ lower peak GPU memory usage. (Under review at MICCAI 2026)

Neuromorphic Processing for Civil Infrastructure Defect Detection

Supervisors: Assoc. Prof. Silvia Tolu, PhD Student Udayanga G.W.K.N. Gamage

Lyngby, Denmark

Sept. 2024 – Jun. 2025

- Labeled and curated drone-based tunnel inspection videos to build defect detection datasets.
- Optimized spiking neural networks (SNN) using the Lava-dl framework by introducing Stage-shared Adaptive LIF, improving inference accuracy with minimal parameter and power overhead.
- Evaluated the proposed model against ANN and SNN models in terms of power consumption, latency, and detection accuracy. (Accepted by Neurocomputing)

PUBLICATIONS

MedDiME: Efficient Latent Diffusion with Adaptive Masking for Medical Counterfactual Generation

Yan Zeng, Changlu Guo, Anders Nymark Christensen, Anders Bjorholm Dahl, Morten Rieger Hannemose
MICCAI 2026, under review

Image-based and Event-based Object Detection on Edge-focused Neuromorphic Hardware: Design, Deployment and Benchmarking

Udayanga G.W.K.N. Gamage, **Yan Zeng**, Cesar Cadena, Matteo Fumagalli, Silvia Tolu
Neurocomputing

WORK EXPERIENCE

Teaching Assistant

Lyngby, Denmark

Course: Distributed Real-Time Systems

Feb. 2026 – Present

- Assisted in exercise sessions and guided students in understanding key concepts of distributed and real-time systems, including task scheduling, synchronization, and system-level design.
- Provided technical support for two course mini-projects involving real-time scheduling analysis (e.g., RM, EDF, and WCRT computation) and Time-Sensitive Networking (TSN) scheduling.
- Helped students design test cases, debug implementations, and interpret experimental results for project assignments.

Teaching Assistant

Lyngby, Denmark

Course: Development Methods for IT Systems

Sep. 2025 – Dec. 2025

- Tutored students in weekly post-lecture exercises and provided Q&A support to help them understand and apply course content.
- Provided support and clarification for group projects, assisting with requirement specifications, system design, and documentation.
- Reviewed and graded project deliverables, including UML diagrams, requirement specifications, and technical reports, providing constructive feedback for improvement.

Data Analysis Intern

Chongqing, China

Yangsheng Tech Group Co., Ltd.

June 10, 2022 – Aug. 27, 2022

- Consolidated raw sales, inventory, and customer feedback data from multiple systems into unified datasets for analysis.
- Performed data cleaning and preprocessing using Python (Pandas, NumPy), including deduplication, missing value imputation, outlier detection, and format standardization.
- Created visual charts of sales trends, inventory fluctuations, and return rates using Matplotlib to support key business metrics tracking.
- Prepared monthly and quarterly analysis reports to inform inventory management and product strategy decisions.

AWARDS AND SCHOLARSHIPS

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| • National Encouragement Scholarship | <i>2020–2021, 2021–2022</i> |
| • Merit Student | <i>2021–2022</i> |
| • Academic Second-Class Scholarship | <i>2021–2022</i> |
| • Academic Third-Class Scholarship | <i>2020–2021, 2022–2023</i> |
| • National “Lanqiao Cup” Algorithm Competition (C++) Second Prize | <i>2021–2022</i> |
| • Outstanding Graduate | <i>2023–2024</i> |

SKILLS

Programming: Python, C++

Deep Learning: PyTorch, Torchvision, MONAI, OpenCV; Generative Models (Diffusion, Flow Matching, Pixel Mean Flow); Spiking Neural Networks (Lava-dl)

Tools & Systems: Git, Linux, HPC (Slurm, LSF; LUMI, Puhti, Mahti), LaTeX